

CARA GIOVANETTI

 caragiovanetti.com  github.com/cgiovanetti

RESEARCH INTERESTS

Beyond-Standard-Model Phenomenology, Big Bang Nucleosynthesis, Computational Methods, Dark Matter and Astroparticle Phenomenology, Particle Cosmology

APPOINTMENTS

University of California, Berkeley, and Lawrence Berkeley National Lab **Berkeley, CA 94720**

Leinweber Institute for Theoretical Physics/Theory Group

Postdoctoral Researcher

Aug. 2025 – Present

EDUCATION

New York University

New York, NY 10003

Center for Cosmology and Particle Physics

Ph.D., Physics

Aug. 2020 – May 2025

Thesis: Cosmology, Astrophysics, and Computation to Advance the
Detection Prospects of Dark Matter

Advisor: Ken Van Tilburg

Princeton University

Princeton, NJ 08544

A.B., Physics

Sep. 2016 – Jun. 2020

Senior Thesis: Cosmological Constraints on Sub-GeV Dark Matter

Advisor: Mariangela Lisanti

FELLOWSHIPS, AWARDS, AND HONORS

- **James Arthur Grant Associate (JAGA) Fellowship, NYU Physics** **AY 2024-2025**
A competitive research assistantship for students working with Center for Cosmology and Particle Physics (CCPP) faculty. Selection reflects CCPP appreciation for the quality and promise of recipient's work.
- **Outstanding Graduate Student Instructor Award, NYU Physics** **2024**
This award recognizes graduate student instructors for their exceptional contribution to teaching some of the department's most challenging and demanding courses.
- **Graduate Research Fellowship, National Science Foundation** **Tenure 2020 – 2023**
Recognizes and supports outstanding graduate students in NSF-supported science, technology, engineering, and mathematics disciplines who are pursuing research-based master's and doctoral degrees at accredited United States institutions.
- **Dean's Doctoral Fellowship, New York University** **2020**
Fellowship awarded to a small number of NYU doctoral students expected to contribute to an intellectually challenging and inclusive educational environment.
- **LeRoy Apker Award Finalist, American Physical Society** **2020**
Recognizes outstanding achievements in physics by undergraduate students.

- Allen G. Shenstone Prize in Physics, *Princeton University* 2019
Given to outstanding Physics undergraduates who have shown excellence in course work and promise in independent research.
- Bell Burnell Fellowship, *Princeton University* 2017, 2018
Summer research fellowship awarded to encourage pursuit of physics.

PUBLICATIONS



8. C. Giovanetti (2026). *A generic ω_b tension in early time solutions to the Hubble tension..* Pre-print. [[arXiv:2604.05095](#)]
7. Z. Zhou, C. Giovanetti, H. Liu (2026). *ABCMB: A Python+JAX Package for the Cosmic Microwave Background Power Spectrum.* Pre-print. [[arXiv:2602.15104](#)]
6. C. Giovanetti, R. Lasenby, K. Van Tilburg (2024). *Orbital Dynamics of the Solar Basin.* **J. High Energ. Phys.** 2024, 7 (2024). [[arXiv:2408.16041](#)]
5. C. Giovanetti, M. Lisanti, H. Liu, S. Mishra-Sharma, J. T. Ruderman (2024). *Cosmological Parameter Estimation with a Joint-Likelihood Analysis of the Cosmic Microwave Background and Big Bang Nucleosynthesis.* **Phys. Rev. D** 112, 063530. [[arXiv:2408.14531](#)]
4. C. Giovanetti, M. Lisanti, H. Liu, S. Mishra-Sharma, J. T. Ruderman (2024). *LINX: A Fast, Differentiable, and Extensible Big Bang Nucleosynthesis Package.* **Phys. Rev. D** 112, 063531. [[arXiv:2408.14538](#)]
3. C. Giovanetti, M. Schmaltz, N. Weiner (2024). *Neutrino-Dark Sector Equilibration and Primordial Element Abundances.* **Phys. Rev. D** 111, 043526. [[arXiv:2402.10264](#)]
2. C. Giovanetti, M. Lisanti, H. Liu, J. T. Ruderman (2021). *Joint CMB and BBN Constraints on Light Dark Sectors with Dark Radiation.* **Phys. Rev. Lett.** 129, 021302. [[arXiv:2109.03246](#)]
1. L. Lancaster, C. Giovanetti, P. Mocz, Y. Kahn, M. Lisanti, D. N. Spergel (2020). *Dynamical Friction in a Fuzzy Dark Matter Universe.* **Journal of Cosmology and Astroparticle Physics**, 2020(01). [[arXiv:1909.06381](#)]

SEMINARS AND INVITED TALKS

- Seminars, 2026
University of California, Davis QMAP Particles/Cosmology Seminar; University of Mainz Theorie Palaver; CERN Cosmology Seminar
- 2025 Gruber Cosmology Prize Symposium New Haven, CT, Oct. 2025
- Seminars, 2025
Sydney Consortium for Particle Physics and Cosmology
- Seminars, 2024
Fermilab Theoretical Physics Seminar; McGill Theory HEP Seminar; University of Washington Particle Theory Seminar; MIT Center for Theoretical Physics Seminar; Stanford Institute for Theoretical Physics Wine and Cheese Seminar; Berkeley Center for Theoretical Physics Particle Seminar; Perimeter Institute Particle Physics Seminar; University of Maryland Elementary Particle Theory Seminar
- Seminars, 2023 and prior
Simons Foundation CCA PACMAN Seminars; Perimeter Institute Particle Physics Seminar; Princeton Pheno & Vino

TEACHING EXPERIENCE

- Quantum Field Theory I, Teaching Assistant Fall 2023
Developed inclusive teaching strategies and active learning techniques to lead one recitation each week.
- Quantum Field Theory I, Grader Fall 2023
- “Quarks to Cosmos”, Laboratory Instructor Fall 2021

SCHOOLS, WORKSHOPS, AND EXTERNAL TRAINING

- JAXtronomy Workshop, Simons Foundation CCA New York, NY
Center for Computational Astrophysics Oct. 2024
- Dark Matter in the Era of Large Surveys, KITP Santa Barbara, CA
Workshop participant, affiliate May – Jun. 2024
- APS JNIPER Fellowship Remote
Joint Network for Informal Physics Education and Research Jan. – May 2024
- APS Science Trust Project Misinformation Workshop Remote
Workshop participant Aug. – Sep. 2023
- APS JNIPER Community Engagement Workshop Remote
Joint Network for Informal Physics Education and Research Jun. – Jul. 2023
Workshop participant
- Lighting New Lampposts for DM and BSM, Stony Brook University Stony Brook, NY
Workshop participant Feb. – Mar. 2023
- TASI, University of Colorado Boulder Boulder, CO
Theoretical Advanced Study Institute in Elementary Particle Physics Jun. – Jul. 2022
Summer Student
- NYU Advanced Science Communication Workshop Series New York, NY
Workshop participant Apr. – May 2022
- NYU Science Communication Workshop Series New York, NY
Workshop participant Sep. – Nov. 2021

INTERNAL ORGANIZING

- *Co-Chair*, NYU CU*iP Organizing Committee 2024 – 2025
Conference for Undergraduate Women and Gender Minorities in Physics
Conference name changed by APS part way through organizing process
- *Chair*, NYU CUWiP Application Committee 2023
Conference for Undergraduate Women in Physics
- *Member*, NYU G-PHORCE Equity and Inclusion Committee 2020 – 2021
Graduate Physics Organization for Research, Culture, and Education
- *Member*, Princeton Physics Climate and Inclusion Committee 2019 – 2020

BROADER IMPACTS AND COMMUNITY ENGAGEMENT

- *Speaker*, [The Lawrence Hall of Science “Champagne Supernova”](#) Berkeley, CA, Nov. 2025
[“Particle Physics? In THIS Economy?”](#)
- *Speaker*, [Skype A Scientist](#) Oct. 2025
- *Author*, [It Takes a Village: Preparing Women in Science for Difficult Times](#) Oct. 2025
[FEBS Letters Women in Science Essay Contest](#), Runner up
- *Scientist Mentor*, [Remote Experience for Young Engineers and Scientists \(REYES\)](#) Jun. – Aug. 2025
Mentored eleven high school and undergraduate students in a hands-on program I developed called *“What’s happening in the Early Universe?”*
- *Panelist*, [New Jersey Governor’s School in the Sciences Career Day](#) Jun. 2025
Careers in Research panel
- *Author*, [Vera Rubin: A Journey from Self-Doubt to Galactic Discoveries](#) Sep. 2024
[APS PhysicsQuest 2010](#), “Force and Motion”
- *Author*, [Why Do So Many Physics Students Want to Work in Academia?](#) Sep. 2024
[APS News](#) “The Back Page”
- *Speaker*, [Liberty Science Center Space Talk](#) Jersey City, NJ, Aug. 2024
[“Dark Matter is Right Next to You”](#)
- *Speaker*, [Astronomy on Tap at Pete’s Candy Store](#) New York, NY, Jun. 2024
[“The Big Bang”](#)
- *Speaker*, [TELL GSAS Sciences \(NYU\)](#) New York, NY, Apr. 2024
[“Dark Matter is Right Next to You”](#)
- *Author*, [Did Neurons Evolve Twice?](#) Jan. 2024
[Scientific American](#)
- *Speaker*, [Cosmic Happy Hour at Book Club](#) New York, NY, Apr. 2023
[“Dark Matter”](#)
- *Guest*, [The Academic Minute Podcast](#) Nov. 2022
[“Dark Matter”](#)
Most listened-to segment of 2022
- *Speaker*, [Marion E. McKeown Elementary School STEM Day](#) Newton, NJ, Oct. 2022
[“Problem Solving with Physics”](#)
- *Guest*, [NSF Discovery Files Podcast](#) Aug. 2022
[“Dark Matter”](#)
- *Scientist Mentor*, [NY Academy of Sciences ASMP](#) New York, NY, Jan. – May 2022
[Afterschool Science Mentorship Program](#)